

HARSH VARDHAN GUPTA

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EDUCATION

Birla Institute of Technology & Science, Pilani

Aug 2019 - Aug 2024

B.E. Computer Science & M.Sc. Economics, GPA: **9.20/10**

Relevant Courses: Operating Systems, Computer Networks, Applied Econometrics, Data Structures & Algorithms, Design & Analysis of Algorithms, DBMS, Deep Learning, Reinforcement Learning, OOPS

SKILLS SUMMARY

Languages/Frameworks: Python, C/C++, SQL, Golang, MongoDB, LLMs, Kubernetes, AWS,

Microservices, Distributed Systems, CI/CD, TensorFlow, Keras

Tools/OS: Docker, Git, MySQL, Linux/UNIX

WORK & RESEARCH EXPERIENCE

Nutanix, Bengaluru

Oct 2024 - Present

Member of Technical Staff - 1

- Working in the Microservices Platform team to transition monolithic services to independently maintainable **containerized workloads** on **Kubernetes**.

- Increased **developer productivity** by **40%** using several pipeline automations in **CI/CD**, **Jenkins & Dolphin Scheduler**, and parallelized the release process.

- Closely working on **kernels** and **Linux gold images** for **Rocky** and **RHEL** distros.

Stack: Python, Kubernetes, Go, AWS, Docker, Linux, CI/CD, Distributed Systems, LLMs.

DevRev, Bengaluru

Jan 2024 - June 2024

Member of Technical Staff Intern

- Designed and implemented a sophisticated **spam management** module for DevRev's tickets and conversations using **Langchain** and advanced **prompt engineering** with **GPT-3.5-turbo**, while creating an add-on automation spam detection module for users that operates in **automatic** and **suggest** modes based on configuration.

- Designed the web elements necessary in **React** for users to showcase spam tickets/conversations.

Stack: Python, Langchain, GenAI, Go, Docker, MongoDB, LLMs.

New York University Tandon School of Engineering, Brooklyn

July 2023 – Dec 2023

Visiting Researcher, supervised by Prof. Nikhil Gupta.

- Determined **grain size** and structure using various **image segmentation** techniques including watershed transformation, **holistically nested edge-based detection**, and granulometry.

- Automating the formerly laborious and error-prone process of manually analyzing microstructures using the **ASTM standardized Heyn-Intercept method**, significantly reducing processing time from **25-30s** to approximately **2s**. The research was funded by the **National Science Foundation (NSF)**.

Stack: Python, OpenCV, Scikit-Learn, TensorFlow, Keras.

G-KnowMe, Bengaluru

Jun 2022 – July 2022

Computer Vision Intern

- Collaborated to develop a web application for digitizing **handwritten oncology forms** using **image processing** and **NLP**, achieving a **40%** reduction in **operational costs** through a format-specific algorithm for enhanced **accuracy** and **efficiency**.
 - Built a lab report digitization module using **OCR**, **dictionary-based chemical identification**, and tracking and **enabling trend analysis** to support clinical decision-making.
- Stack: Python, OpenCV, Scikit-Learn, TensorFlow, Keras.

PUBLICATIONS & CONFERENCES

Presented research paper titled "**Media Sentiment and IPO Underpricing in Indian Markets**" at the prestigious **ICBT'24 conference** at **Oxford University**. This same research was subsequently honored with the **Best Paper Award** at the **BAMS-ORSI 2024** Conference hosted by **IIT Bombay**, recognizing its significant contribution to the intersection of **Finance** and **AI**. {Ongoing publication}

Gupta, H. V., Chen, G. L., Das, A., Sharma, A., Campbell, A., Murray, P. and Gupta, N. Automated Particle Size Measurement in Foundry Applications: A Study of Imaging Techniques. 2024, Indian Foundry Journal. Vol. 70 (11) pp. TBD. {**Best Paper Award Recipient**}

Gupta, H. V., Chen, G. L., Das, A., Sharma, A., Campbell, A., Murray, P. and Gupta, N. Image Analysis Methods for Grain Size Analysis: An Overview and A Case Study. 2024, Indian Foundry Journal. Vol. 70 (3) pp. TBD.

PROJECTS

Deep Reinforcement Learning for Automated Stock Trading

Feb 2023 - May 2023

Python, FinRL, OpenAI Gym, TensorFlow

- Designed a reinforcement learning bot for the **DOW Jones** equity index using an **ensemble strategy** of PPO, A2C, and DDPG algorithms to maximize the **Sharpe ratio**.
- Extended this approach to the **NIFTY 50** index with **fine-tuned hyperparameters** and conducted **ablation studies** including **optimizers**, **time steps**, and **discount factors**, to outperform the market. Conducted thorough **backtesting** across both markets to validate performance.

Knight Tour Problem using Multithreading Approach

Oct 2022 - Dec 2022

C, POSIX, Ubuntu

- Solved the **Knight's Tour** Problem for an NxN chessboard for a given starting position using **multithreading** and **synchronization** by mutex locks. Also implemented the **Warnsdorff's** heuristic for optimality and execution time.

Alzheimer's Disease Stage Diagnosis on MRI Brain Scans

Oct 2022 - July 2022

Python, PyTorch, Keras, TensorFlow, OpenCV

- Applied preprocessing techniques such as **skull stripping**, followed by **deep learning** and **machine learning** methods including **autoencoders**, **SVM**, **3D CNNs**, and **transfer learning** on axial, coronal, and sagittal MRI slices to classify subjects into **CN** (cognitively normal), **MCI** (mild cognitive impairment), and **AD** (Alzheimer's Disease) stages.

TEACHING & RESEARCH EXPERIENCE

FIN F412: Security Analysis & Portfolio Mgmt.

Jan 2023 - May 2023

Teaching Assistant, Dept. of Economics and Finance

- Instructor: Prof. Shreya Biswas, Spring 2023
- Assisting the course instructor in tutorials, preparation and validation of exercises and solutions, and operating the website/CMS.

FIN F315: Financial Mgmt.

Sept 2022 - Dec 2023

Teaching Assistant, Dept. of Economics and Finance

- Instructor: Prof. Nivedita Sinha, Fall 2022
- Designed **Python** and **Excel-based exercises** involving ratio analysis, financial forecasting, and valuation modeling. Assisted students in **automating valuation templates** and **analyzing corporate financial** statements using Python, Excel, and Pandas.

A Study of Ranking of Higher Institutions in India

Sept 2022 - May 2023

Research Assistant, Dept. of Economics and Finance

- Jointly Supervised by Prof. Nivedita Sinha and Prof. Vivekananda Mukherjee.
- Involved with **data extraction, cleaning, and analyzing results** for the ongoing research project in the department funded by BITS Pilani. Conducted statistical analysis of **NIRF rankings** and **media sentiment** data to identify key drivers of institutional performance and their impact on **student decision-making**.

SCHOLARSHIPS & AWARDS

Awarded the **Merit Scholarship twice**, which is given to the **Top 2%** (CGPA) students out of **1100**. 2023-2024

EXTRA-CURRICULAR & SOCIETY MEMBERSHIPS

- Chess player with **1805 Rapid** Rating in Lichess equivalent to **~1400 FIDE** Rating. 2024
- Member of **Department & Sponsorship & Marketing** responsible for bringing sponsors and collecting funds for college events. 2019-2021
- Member of **Department of Recreational Activities** responsible for organizing fun events like Treasure Hunt, board games and trivia nights for a pool of 1000 students. 2019-2021
- **Grade 6 piano player** as per **ABRSM** (Associated Board of the Royal Schools of Music) grading system. 2006-2012
- Graduated with **UCMAS Abacus degree** with **State Rank 5** in Rajasthan, India and **City Champion** title. 2011

REFERENCES

Prof. Nikhil Gupta

Professor

Mechanical and Aerospace Engineering Department

Civil and Urban Engineering Department

NYU Center for Cybersecurity

Relationship: Undergrad Thesis Advisor

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Prof. Nivedita Sinha

Assistant Professor

Department of Economics and Finance

Relationship: Co-Author

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